Polarity

Inverters are useful buffers but flip the signal; odd counts need a fix

The idea

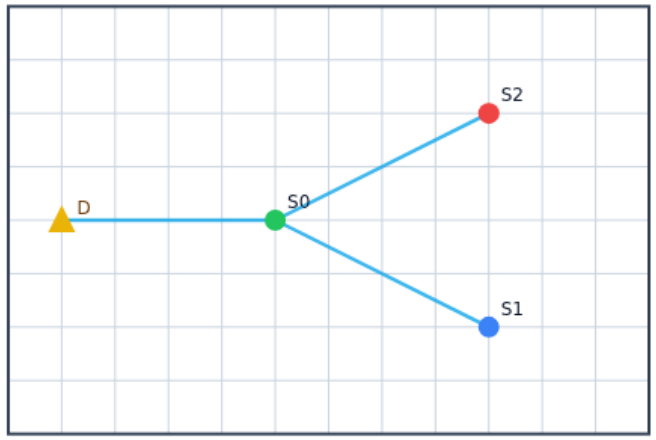
- Each INV_X1 flips polarity
- Polarity_fix reports flips and whether the path is still positive

Invert flag

POLARITY

Invert flag

Step 1 / 5 · flag · module03-01-polarity-invert



Explanation

INV_X1 has invert=true in the library.

- Library

```
Purple
```

Invert flag

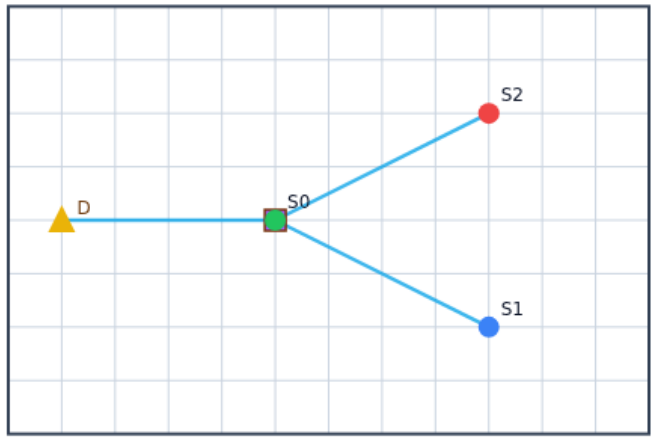
One flip

POLARITY

One flip

Step 2 / 5 · one · module03-01-polarity-invert

flips=1



Explanation

A single INV yields negative polarity.

- flips=1

```
positive=false
```

One flip

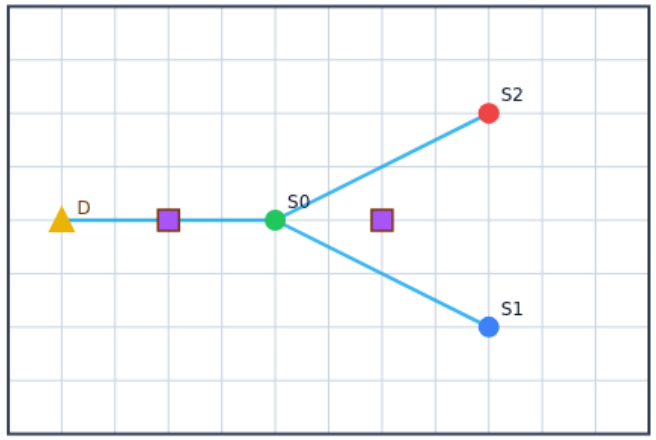
Two flips

POLARITY

Two flips

Step 3 / 5 · two · module03-01-polarity-invert

flips=2



Explanation

Two INVs restore positive sense.

- Parity

```
positive=true
```

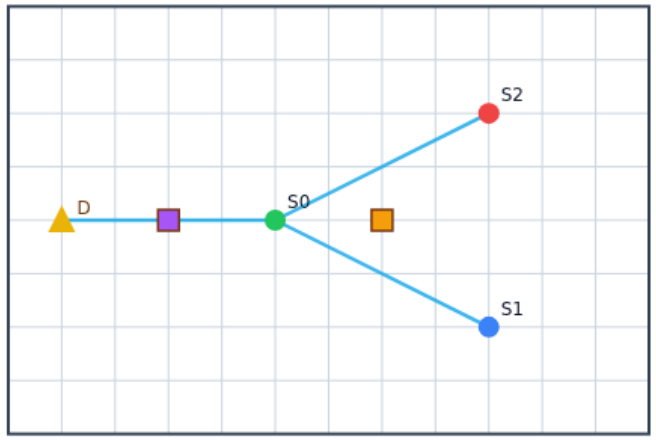
Two flips

Mix with BUF

POLARITY

Mix with BUF

Step 4 / 5 · mix · module03-01-polarity-invert



Explanation

BUF cells do not flip.

- BUF_X1

```
Parity unchanged
```

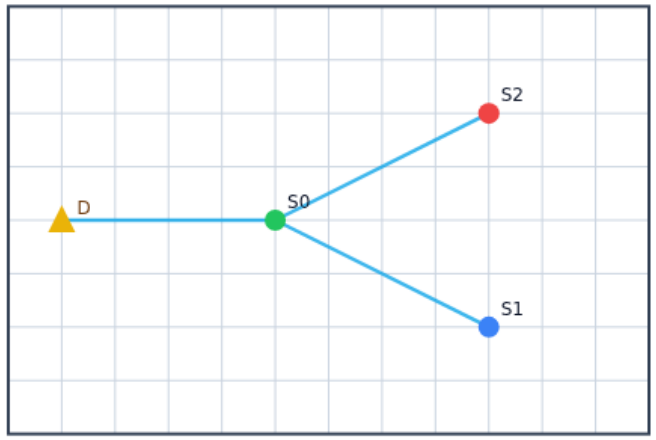
Mix with BUF

Design rule

POLARITY

Design rule

Step 5 / 5 · rule · module03-01-polarity-invert



Explanation

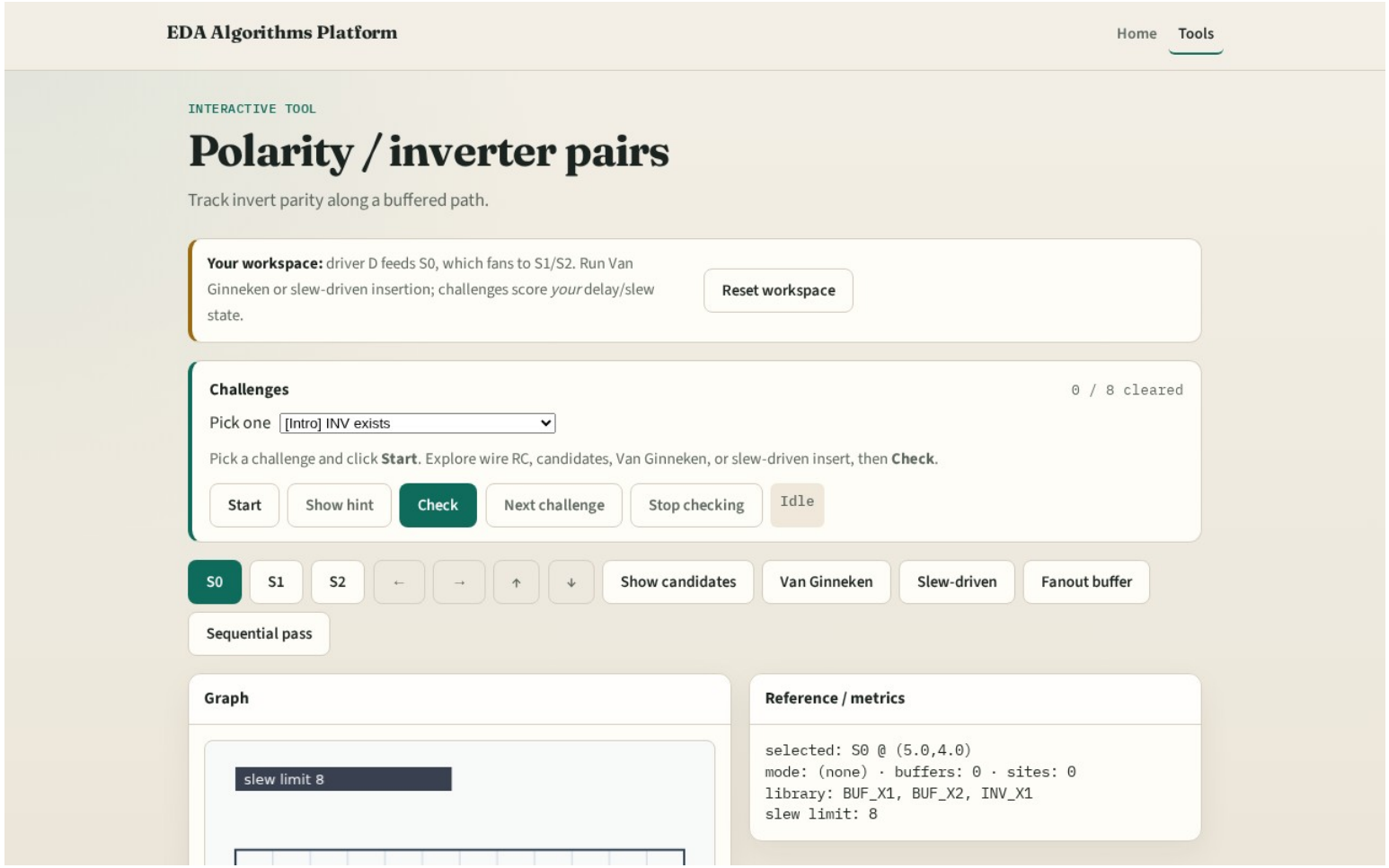
Useful-skew flows may allow phase; this toy tracks parity.

- Toy scope

```
Fanout next
```

Design rule

Browser lab track



Browser lab starter

Implement track

- Call `polarity_fix([INV],[BUF])`
- Assert flips and positive flags

Pitfalls

- Counting BUF as invert
- Forgetting parity
- Using INV in VG when allow_invert is false

Your turn

- Next: fanout buffering at the junction